MIL-B-15228F
30 April 1982
SUPERSEDING
MIL-B-15228E
25 October 1967

MILITARY SPECIFICATION
BED, BUNK, STEEL, ROUND TUBE, SINGLE AND DOUBLE-DECK (CONVERTIBLE TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for component materials and their assembly into a single bunk bed with adapters for conversion to double-deck bunk beds.

1.2 Classification. Bunk beds shall be of one type and shall be of the sizes shown on figure 1 and as specified below (see 6.2).

Size 1 - 36-inches wide by 79-inches long overall.

Size 2 - 36-inches wide by 84-inches long overall.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Facilities Engineering Command, Code 0432, 200 Stovall Street, Alexandria, VA 22332 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7105

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
TT-P-320	- Pigment, Aluminum; Powder and Paste, for Paint.
TT-P-659	- Primer-Coating and Surfacer; Synthetic, Tints and White, (for Metal and Wood Surfaces).
TT-P-1757	- Primer, Coating, Zinc Chromate, Low Moisture Sensitivity.
TT-T-291	- Thinner; Paint Volatile Spirits (Petroleum-Spirits).
TT-V-109	- Varnish, Spar Alkyd-Resin.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED- STD-151 Supp. A and B	- Metals, Test Methods.
FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A36	- Structural Steel
ASTM A 366	- Steel, Carbon, Cold-Rolled Sheet, Commercial Quality
ASTM A513	- Eletric-Resistance Welded Carbon and Alloy Steel Mechanical Tubing
ASTM A569	- Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, N.Y. 10017).

American Welding Society, Inc. (AWI)

AWS A3.0	- Terms and Definitions

(Application for copies should be addressed to the American Welding Society, Inc., 2501 Northwest 7th Street, Miami, FL 33125.)

Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The bunk bed shall be of all steel three-piece construction consisting of matching round steel tube, head and foot ends, a wedge type corner lock with four bearing points, and a fully-framed spring unit with sinuous (no-sag type) spring fabric, as shown on figures 2 and 3. Bunk beds shall be of two sizes as specified in 1.2.

3.2 Identical items. All units of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations shall be acceptable without prior written approval of the contracting officer.

*3.3 First article. Unless otherwise specified (see 6.2), the contractor shall furnish one bunk bed, two beds if adapters are included in the order, for first article inspection and approval (see 4.2.1 and 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

*3.4 Materials. Use materials free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Material not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be fabricated from recovered raw materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials reprocessed to become a source of raw materials, as opposed to virgin raw materials which include previously unused metal or metal ore, any undeveloped resource, or new technology which will become source of raw material. Interpret none of the above to mean that the use of used or rebuilt products are allowed, unless otherwise specified.

3.4.1 Steel. All steel shall be smooth, free from rust, scale, pits, scratches, laps and buckles affecting strength and appearance. Unless otherwise specified, the gage numbers or thicknesses specified for steel are subject to standard tolerances. Steel shall be tested in accordance with 4.5.1 where applicable.

3.4.1.1 Steel tubing. Steel tubing for bed ends, adapters, and stretcher braces shall be in accordance with ASTM A513, electric resistance welded type, round.

3.4.1.2 Steel angles. Steel angles for spring fabric frame shall be in accordance with ASTM A36 and shall be 2 by 1-1/2 by 3/16-inches thick. Edges of angles shall be slightly rounded to avoid sharpness.

3.4.1.3 Spring fabric. Spring fabric shall be high carbon sinuous (no-sag type) furniture spring wire in accordance with Table 7-55 AISI Steel Products Manual. Spring wire shall be No. 10 US Steel Wire Gage, 31-inches long in the flat position, containing 19 full loops including the "Z" hook ends. Spring fabric shall be fabricated to 7-1/2-inch diameter plus or minus 1/2 inch. Cut ends shall not have burrs or sharp projections.

3.4.1.4 Spring fabric links. Spring fabric links shall be No. 11 US Steel Wire Gage basic steel wire in accordance with Table 7-48 AISI Steel Products Manual with galvanized finish. Dimensional and size requirements shall be in accordance with figure 4.

3.4.1.5 Helical springs. Helical springs shall be No. 12 US Steel Wire Gage steel premier wire with electrodeposited zinc coating. Tensile strength range in PSI shall be 204,000 to 230,000. Dimensional requirements shall be in accordance with figure 4.

3.4.1.6 Corner locks.

*3.4.1.6.1 Male corner locks. The male corner locks shall be in accordance with ASTM A569, hot-rolled sheet or strip, commercial quality steel, pickled and oiled of shape, thickness, and dimensional requirements shown on figure 5. As an alternate, the male corner locks shall be in accordance with ASTM A366, cold-rolled sheet, commercial quality, and oiled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

3.4.1.6.2 Male corner lock studs. Male corner lock studs shall be AISI Designation C-1010 steel. When upset, shank shall be annealed for riveting with a maximum Rockwell hardness of B-40. When machined, studs shall be AISI Designation CF-B-1113, C-1213, or C-1215 steel. Dimensional requirements shall be as shown figure 5.

3.4.1.6.3 Female corner locks. The female corner locks shall be in accordance with the requirements specified in 3.4.1.6.3 except that the shape, thickness, and dimensional requirements shall be as shown on figure 6.

3.4.2 Primer-surfacer. Primer-surfacer shall conform to TT-P-659.

*3.4.3 Enamel. Enamel shall be in accordance with TT-E-529, Type II. Color shall be in accordance with FED-STD-595 Nos. 16251, (Gray gloss), 20059, 20227, 23578, or 27038 as specified (see 3.7.3.1 and 6.2).

*3.4.4 Zinc-chromate primer. Zinc-chromate primer shall conform to TT-P-1757.

3.4.5 Aluminum enamel. Aluminum enamel shall consist of one pound of aluminum paste-pigment conforming to type II, class A of TT-P-320 reduced with one pint of mineral spirits conforming to grade 1 of TT-T-291 and then mixed with one gallon of alkyd spar varnish conforming to TT-V-109.

*3.5 Accessories.

3.5.1 Steel caps. Steel caps for top of bed end posts shall be not less than No. 24 US Standard Gage carbon steel, hardened and acid dipped for paint adhesion. Caps shall have 12 prongs with dimensional requirements shown on part A of figure 7.

3.5.2 Bed shoes. Steel bed shoes for bed end posts shall be round body die-formed or machined from solid AISI Designation CF-B-1113 steel, chromium plated. Dimensional and hardening requirements shall be in accordance with part C of figure 7. The snap ring for securing steel shoe in bottom of the end post shall be in accordance with requirements of part D of figure 7. As an alternate, bed shoes in accordance with the requirements of part B of figure 7 may be provided. The shoe shall be a "file hard", nickel plated, cold rolled steel, die-formed base curled over a zinc plated hot rolled, die-formed inside shell with the resultant gap between the formed diameter of the inside shell and the curled base not to exceed 1/32 inch. The spring retainer shall be AISI Designation C-1010 cold rolled steel with a Rockwell hardness of B-85 to B-95 and assembled to the bed shoe with a zinc plated bright basic steel wire rivet and a zinc plated hot rolled steel washer between the spring retainer and the inside shell to absorb side thrust when mounted in the bed end post.

3.5.3 Adapter insert. Insert for adapter shall be cold rolled, sheet steel, matte finish, in accordance with ASTM A366 fabricated into open joint tubing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

3.6 General construction. To insure interchangeability, beds shall be similar and of same construction in accordance with design, material thickness, dimensions and tolerances shown in figure 1 through 13 and specified herein. Materials required shall be in accordance with details specified in 3.4 and accessories specified in 3.5. Components for both size beds shall be the same, except the 84-inch long bed (Size 2) shall require two additional strands of spring fabric, side rails shall be 5-inches longer than the 79-inch long bed (Size 1), and spring fabric link sizes shall be in accordance with figure 4.

*3.6.1 Bed ends. Head and foot ends shall be of same construction as shown on figure 8. Each bed end shall consist of two main posts, two horizontal tubes, four intermediate vertical tubes, two female corner locks, two steel caps and two bed shoes. Main posts shall be round steel tube, 28-1/2-inches high exclusive of shoes and caps, 1-1/2-inch outside diameter with a wall thickness of 0.083 inch. Horizontal tubes shall be steel, round, 7/8-inch outside diameter with a wall thickness of 0.065 inch. Intermediate tubes (filler) shall be steel, round, 1/2-inch outside diameter with a wall thickness of 0.065 inch. Horizontal tubes shall extend between the posts and intermediate tubes in a vertical position, parallel to the post tubes as shown on figure 8. All tube connections shall be made by inserting them into connecting tubes, placing all components in jigs and joining by most applicable methods of welding around the entire perimeter of joints (see 3.9.1). Posts shall be closed at top with steel caps requiring 5-pound pull plus or minus 1-pound to remove the cap and fitted at bottom with steel shoes requiring 10-pound pull plus or minus 1-pound to remove from the post. The shoes shall be removeable without use of tools, so beds can be converted double-deck type when desired. Female corner lock shall be placed on main post as shown on figure 8 to match the male corner lock on the spring unit. The female corner lock shall be properly jigged and welded across the top, bottom, down one inch from top and up one inch from bottom on the unexposed side as shown on figure 6.

3.6.2 Bed spring unit. The bed spring unit shall consist of an all welded angle frame, sinuous (no-sag type) spring fabric, two round tubular steel stretcher braces, and four male corner locks as shown on figures 2 and 3. The frame shall be constructed of steel angle sides and ends in accordance with 3.4.1.2 herein, with the 2-inch flange in the vertical position and the 1-1/2-inch horizontal flanges punched in accordance with figure 9 for insertion of spring fabric "Z" hook ends in side angles and spring helicals in the end angels. The horizontal flanges of the end angles shall overlay the horizontal flanges of the side angles with width and length as shown on figures 2 and 3. Frame shall be properly jigged and welded (see 3.9.1) as shown on figures 10 and 11. Two stretcher braces of steel tubing with 7/8-inch outside diameter and a 0.065 inch wall thickness shall be bowed, and the ends flattened as shown on figure 12. The stretcher braces shall be located on side angles as shown on figure 2 for Size 1 bed and as shown on figure 3 for Size 2 bed. Each end of the stretcher brace shall be secured to horizontal flange of side angle by a continuous weld along the entire union of the side angle surface and the stretcher brace edge. The male corner lock as shown on figure 5 shall be of the double stud design. The male corner lock plate shall be provided with two shouldered steel studs riveted (see 3.9.2) thereto to engage the female corner lock as shown on figure 5. The male

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

corner lock shall be located on side angle as shown on figures 2 or 3 and welded thereto as on shown figure 10. The upper and lower extension on the male lock end shall butt against the slotted surface of the female lock providing a lock wedge fit to prevent the bed from racking. There shall be four point contact between male and female corner locks; at upper and lower extensions on male lock end and the two studs. The spring fabric shall consist of 18 equally spaced transverse strands of no-sag type spring fabric for Size 1 bed and 20 strands for Size 2 bed. Each strand shall be connected to the adjacent strand with 5 rows of links (figure 4) clinched over the spring fabric. The "Z" hook ends of each strand shall be inserted in holes in horizontal flange of each side angle in a manner to provide secure retension; free from scratching and tearing hazard. Each end strand shall be attached to the end angle with five helical springs (figure 4).

3.6.3 Adapters. When specified (see 6.2), each pair of bunk beds shall be supplied with a set of four adapters for converting to a double-deck bed. Adapters shall consist of a straight section of steel tubing 10-1/2-inches long with an outside diameter of 1-1/2-inches and a wall thickness of 0.083 inch and inserts (inner sleeves) which shall project 3 inches from each end as shown on figure 13. Insert shall be a drive fit within the outer tube and shall be further secured with two drive screws. Outer edges of insert ends shall be beveled to ease insertion into bed posts. Inserts shall be adjusted as necessary to fit snugly, after painting, into top of lower bed and bottom of upper bed posts when caps and shoes are removed. Adapters shall be examined for assembly and disassembly in accordance with 4.4.2.

*3.7 Treatment and painting. All surfaces of the bunk beds, except plated steel shoes, shall be painted. Application of primer and finish shall produce smooth and uniform surfaces without runs, wrinkles, and free from objectionable defects such as scale under the finish, rust or discoloring.

3.7.1 Surface preparation. All surfaces to be painted shall be treated in accordance with type I or type II of TT-C-490 except phosphate coating may be applied in three stages and a non-chromated final acidic rinse maybe used.

3.7.2 Priming.

*3.7.2.1 Primer-surfacer. All treated surfaces, except bed spring, shall be given one coat of primer-surfacer applied to a minimum dry film thickness of 0.5 mil. The primer-surfacer shall be baked for not less than 20 minutes at a temperature of 300 degrees F, or for an equivalent baking period as recommended by the paint manufacturer.

3.7.2.2 Zinc-chromate primer. The treated surfaces of bed spring units shall be given one coat of zinc-chromate primer applied to a minimum dry film thickness of 0.5 mil.

3.7.3 Finish.

*3.7.3.1 Enamel. All surfaces primed with primer-surfacer shall be finished with a coat of enamel of color specified (see 3.4.3 and 6.2) applied

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

to a minimum dry film thickness of 0.6 mil. The enamel shall be baked for not less than 20 minutes at 300 degrees F, or for an equivalent baking period at a higher temperature as recommended by the paint manufacturer.

3.7.3.2 Aluminum enamel. All surfaces primed with zinc-chromate primer shall be finished with a coat of aluminum enamel applied to a minumum dry film thickness of 0.8 mil. The aluminum enamel shall be baked for not less than 2 hours at 180 degrees F, or for an equivalent baking period as recommended by the varnish manufacturer, except that baking temperature not exceed 325 degrees F.

3.8 Marking. The letters "US" in characters 1-inch high shall be indented at center of horizontal or vertical surface on one end angle of each bed. The specification number, stock number,, nomenclature, manufacturer's name, trade name or trademark of such character easily identifiable with the manufacturer, and the year and contract number shall be indented or on an aluminum plate permanently attached on same end angle surface and to right of indented characters "US". Bed ends manufacturer shall be identified by identing the manufacturer's initials or trademark of such character easily identifiable with the manufacturer on one surface of one or both female corner locks. Letters or trade mark shall be approximately 1/2-inch high at location shown on figure 6.

3.9 Workmanship. Workmanship shall be of the highest quality throughout and in accordance with the best standard practice used in the manufacturer of high grade commercial furniture of this type.

3.9.1 Welding. All welds shall be sound, and smooth, free from pits, holes, and fissures. Bed spring fillet welds shall be in accordance with the definition and size requirements of AWI A3.0. The deposited weld metal shall be thoroughly fused with base metal and edges of union and shall penetrate into the root of the joint. All rough surfaces or projecting fins shall be ground smooth to prevent personal injury or damage to bedding. Weld spatter or flux shall be removed from surfaces surrounding the welding area.

3.9.2 Riveting. Studs shall be secured to the male corner locks by peening as shown on figure 5. The studs shall be tight with the shoulder and end in full contact with the male corner lock surfaces. The head of the stud end shall be flat, parallel to the male corner lock surface, and edges free of burrs or fins.

3.10 Assembly instruction. Prior to shipment, each shipping container (carton) shall contain assembly instructions for single bed, double decking (see 6.4). In addition, a sketch similar to figure 14 shall be included in instructions to preclude the possibility of forcing male corner lock studs to seat or bottom in female corner lock. Instructions shall be provided on a white paper sheet 8-inches wide by 10-1/2-inches long.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (4.2.1).
(b) Quality conformance inspection (4.2.2).

4.2.1 First article inspection. First article inspection shall be performed when first article samples are required (see 3.3). This inspection shall include the examination of 4.4 and tests of 4.5.

4.2.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.4 and 4.6, also the tests of 4.5.

*4.3 Sampling. Sampling shall be in accordance with MIL-STD-105. A lot shall consist of one day's production of bunkbeds or adapters, but shall not consist of not more than 500 bunk beds. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be inspected using a tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

*4.3.1 Sampling for examination. Examination shall be based on inspection level S-4 and an acceptable quality level of 2.5 defects per hundred units (DPU) for major defects and 4.0 DPU for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-4 and an acceptable quality level of 2.5 DPU.

4.4 Examination. Each random sample shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table I, 4.4.1,, and 4.4.2. Any sample having one or more of the defects listed, as applicable, shall be considered a defective unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

*TABLE I. Classification of defects.

Categories	Defects	Requirement
Critical	None defined.	
Major	(In process examination)	
101	Surfaces to be painted not prepared as specified.	3.7.1
102	Primer-surfacer on bed ends not as specified.	3.7.2.1
103	Zinc-chromate primer on spring unit not as specified.	3.7.2.2
104	Enamel on bed end, baking cycle not as specified.	3.7.3.1
104	Aluminum enamel on spring unit, baking cycle not as specified.	3.7.3.2
Major	(End item examination)	
105	Design and construction not as specified.	3.1
106	Overall dimensions not as specified.	fig. 1
107	Parts not manufactured to interchangeable standards.	3.2 and 3.6
108	Steel surfaces not as specified.	3.4.1
109	Steel components not as specified.	3.4.1.1 thru 3.4.1.6
110	Primer-surfacer not as specified.	3.4.2
111	Enamel not as specified.	3.4.3
112	Zinc-chromate primer not as specified.	3.4.4
113	Aluminum enamel not as specified.	3.4.5
114	Steel caps not as specified.	3.5.1
115	Bed shoes not as specified.	3.5.2
116	Adapters not as specified.	3.5.3
117	Bed end dimensions not as specified.	3.6.1 and fig. 8
118	Wall thickness of main posts not as specified.	3.6.1
119	Wall thickness of horizontal and intermediate vertical tubes not as specifed.	3.6.1
120	Tube connections not as specified.	3.6.1
121	Shoes not removable without use of tools.	3.6.1
122	Female corner locks not welded as specified.	3.6.1 and fig. 6
123	Bed spring unit angle frame members not welded as specified.	3.6.2 and figs. 10 and 11
124	Long leg of angle sides not in vertical position as specified.	3.6.2
125	Stretcher braces not as specified.	3.6.2
126	Stretcher braces not located as specified.	3.6.2
127	Corner locks not located as specified.	3.6.2
128	Number of strands of spring fabric not as specified.	3.6.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

129	Spring fabric loose and not attached as specified.	3.6.2
130	Hook ends of spring fabric not free of scratching and tearing hazard.	3.6.2
131	When adapters are specified, four adapters not provided for each set of two beds.	3.6.3
132	Dimensions and thicknesses for adapters not as specified.	3.6.3
133	Outer edges of adapter insert ends not beveled.	3.6.3
134	Finish not without runs, wrinkles, or areas of thin or no film, or other defects such as indication of rust or scale under finish or discoloring.	3.7
135	Primers not of thicknesses specified.	3.7.2
136	Enamels not of thicknesses specified.	3.7.3
137	Welding not as specified.	3.9.1
138	Studs not secured as specified.	3.9.2
139	Assembly instructions not provided.	3.9.10
Minor:		
201	Edges of angles not rounded.	3.4.1.2
202	Color not as specified.	3.4.3
203	Pull necessary to remove steel caps not as specified.	3.6.1
204	Pull necessary to remove steel shoes not as specified.	3.6.1
205	Identification marking missing.	3.8

4.4.1 Bed assembly examination. Sample beds from each inspection lot shall be assembled to determine ease of assembly and proper fit of mating parts. In addition, the beds shall double-decked to assure proper overall dimensions among beds. Failure of studs and male corner lock extensions (see 3.6.2) to fit properly, requiring excessive force or use of tools to assemble or inability to match top bed post bottoms and bottom bed top when double-decking each shall consititute a major defect.

4.4.2 Adapter assembly examination. When adapters are procured separately or with beds, samples shall be examined for ease of double-decking. Assembly and disassembly with beds or steel tubes of same diameter and wall thickness as bed main posts as specified shall be performed manually without the use of tools and shall be a snug fit. Inside diameter of bed posts or steel tubes shall be checked periodically for proper dimensions (see figure 13).

4.5 Tests.

4.5.1 Metals. Specimens of metal components shall be tested in accordance with FED-STD-151, Supp. A and B, wherever applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

4.5.2 Bed test. A 600 pound load shall be placed across the side angles of spring unit of an assembled bed, midway between the bed ends. The test load bottom or base shall be approximately 18-inches wide. At the center of the spring unit, the deflection of the spring unit side angles shall be not more than 7/16 inch with the 600 pound load in place. Upon removal of the test load, the spring unit shall return to its original position with no visible distortion in the bed locks or spring unit frame.

*4.6 Packing inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of Section 5. The inspection shall be in accordance with MIL-STD-105. A completed pack prepared for shipment shall be considered a unit of product. A lot shall consist of all completed packs prepared for shipment during any one day period. The inspection level shall be level II with an AQL of 4.0 percent defective.

5. PACKAGING

*5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The main posts and the horizontal tubes of the bed ends (head and foot ends), and the end angles and corner locks of the bed spring unit shall be completely wrapped with cushioning material securely fastened in place with tape or twine in a manner which will prevent shifting or becoming detached. Additional cushioning material shall be used as required to protect any other metal surfaces from damage. Two complete single bunk beds shall be arranged into a compact assembly in order to provide the smallest practicable shipping cube and to provide support for the faces and edges of the exterior shipping container, as follows: The four cushioned bed ends shall be placed inside one inverted bed spring unit in the following manner. One bed end shall be placed with the corner locks facing upward and one main post resting against the inside of the bed spring unit end angle and with the horizontal tubes parallel to and equidistant from the unit side angles. A second bed end shall be placed in the same position on top of the first bed end except that the corner locks shall face downward, one main post shall rest against the outside of the corner lock of the first bed end, and the horizontal tubes shall be offset from the horizontal tubes of the first bed end by a distance equal to about the diameter of one tube. The other two bed ends shall be placed at the other end of the inverted bed spring unit in the same manner. The second bed spring unit with the spring fabric facing outward shall then be placed on top and nested and offset so that all corner locks are adjacent to and alongside the corresponding corner locks of the inverted unit; at each corner, the corner lock of one unit bears on the angle of the other unit. The assembly shall be secured in this position by the applcation of two flat steel straps at each end which completely encircle the main posts and the end angles. Each strap shall be located between the horizontal tube and the side angle and shall be drawn as tight as possible to provide a compact non-shifting assembly. The strapping shall be minimum 3/4-inch width and shall be prevented from coming in direct contact with metal surfaces of the assembly by means of the applied cushioning

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

or by use of additional separating material. When adpaters are furnished with the bunk beds (see 3.6.3), they shall be cushioned with the materials specified herein and secured in an appropriate location inside the bed spring units with filament reinforced tape or flat steel strapping.

5.1.1.1 Adapters (procured separately). Adapters procured separately from bunk beds shall be packaged in quantities of four in a close-fitting fiberboard box of minimum 175 psi bursting strength and the box closed and secured. The adapters shall be cushioned to prevent contact between pieces.

5.1.2 Level C. Bunk beds (with adapters when furnished) and adapters (when furnished separately) shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. Two complete single bunk beds, packaged as specified in 5.1.1, shall be packed in a close-fitting box conforming to PPP-B-601 overseas type or PPP-B-621, class 2. The box shall be close-fitting such that the cushioning material on the outermost parts of the assembly bears against the inner surfaces of the exterior shipping container in a snug manner. The box shall be provided with a waterproof case liner conforming to MIL-L-10547, sealed in accordance with the appendix thereto.

5.2.1.1 Adapters (procured separately). Adapters procured seaprately from bunk beds, packaged as specified in 5.1.1.1, shall be packed in boxes specified in 5.2.1. The boxes shall be provided with case liners as specified in 5.2.1.

5.2.2 Level B. Two complete single bunk beds, packaged as specified in 5.1.1, shall be packed in a close-fitting box conforming to either PPP-B-640, class 2, PPP-B-601, domestic type, or PPP-B-621, class 1. PPP-B-640 boxes shall be closured with flat steel strapping and waterproofed with tape in accordance with the appendix to the box specification.

5.2.2.1 Adapters (procured separately). Adapters procured separately and packaged as specified in 5.1.1.1 shall be packed as specified in 5.2.2 or in boxes conforming to PPP-B-636, class weather-resistant and closed in accordance with Method V of the appendix thereto.

5.2.3 Commercial. Packaged bunk beds (with adapters when furnished) and packaged adapters (when furnished separately) shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Boxes containing both bunk beds and adapters shall be marked noting that adapters are also included in the box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. This specification covers bunk beds for use in unaccompanied enlisted personnel housing.

*6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Whether first article is required (see 3.3).
d. Color, required (see 3.4.3).
e. Whether each pair of single beds shall be supplied with a set of four adapters (see 3.6.3).
f. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Nomenclature. The approved nomenclature of the item covered by this specification is as follows:

Bed, Bunk, Steel, Round Tube, Single and Double-Deck.

6.4 Bunk bed assembly instructions.

Single bed assembly instructions. Assemble a single bunk bed by placing a spring unit on the floor with spring up, stand a bed end at each end of the spring unit, with one person standing outside of each bed end, lift the spring unit up and place between the slotted extensions on the main posts of the bed end and fit the studs into to the slots. Lightly tap each corner to set the studs in the slots. DO NOT FORCE THE STUDS TO SEAT AT BOTTOM OF THE SLOTS. See illustration below.

Double decking instructions. Assemble two bunk beds as instructed above. Remove the caps from the top of the main posts at each corner of one bed and remove the bed shoes from the bottom of the main posts of the other bed. Place an adapter into each post from which the caps were removed. Lift the other bed above the adapters and insert the top of the adapters into the bottom of the main posts.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of one bed or two if adapters are specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.6 Changes from previous issue. Paragraphs of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

Custodians:
Army - GL
Navy - YD
Air Force - 99

Review Activities:

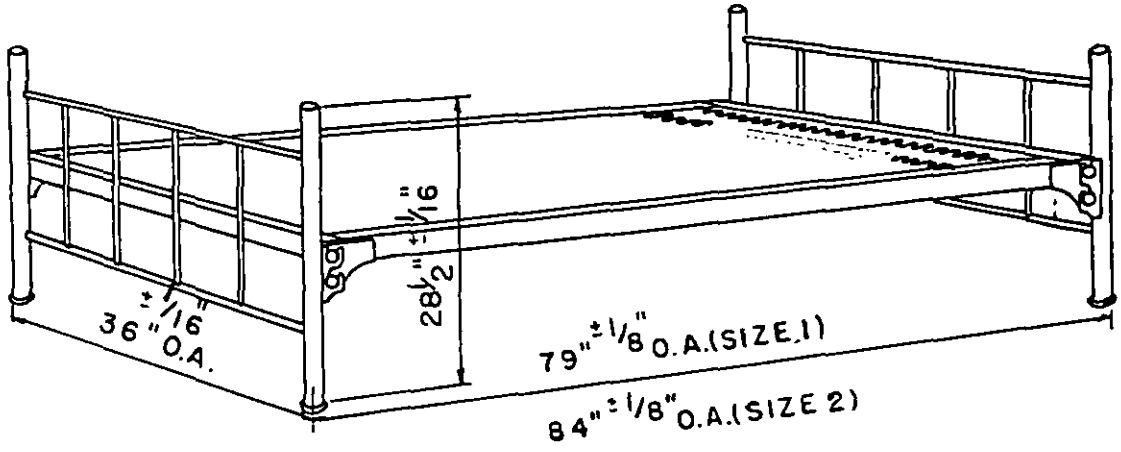
Army - MD
Navy - MC

Preparing Activity
Navy - YD

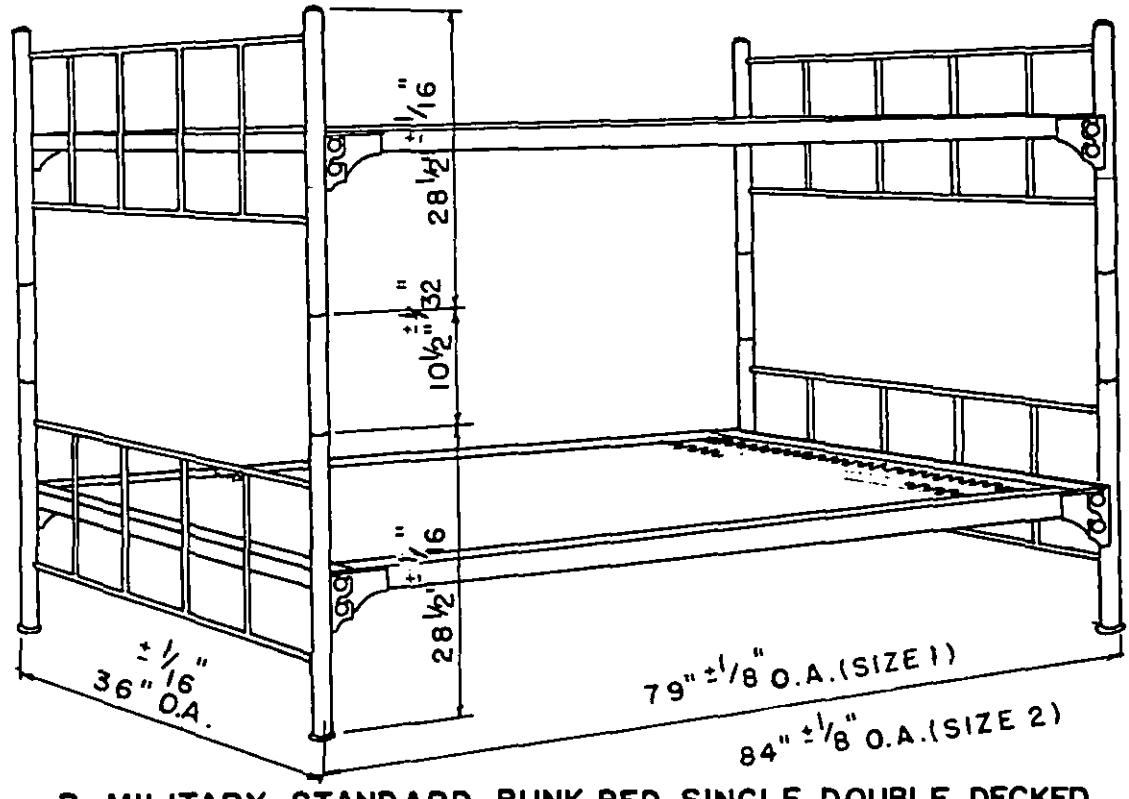
Project No. 7105 - 0240

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

NOTE: MEASURE OVERALL BUNK BED LENGTH AT MID POINT BETWEEN TOP AND BOTTOM OF BED POSTS.



A. MILITARY STANDARD BUNK BED, SINGLE



B. MILITARY STANDARD BUNK BED, SINGLE DOUBLE DECKED

FIGURE 1. MILITARY STANDARD BUNK BED, STEEL, SINGLE AND DOUBLE-DECK (CONVERTIBLE TYPE)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

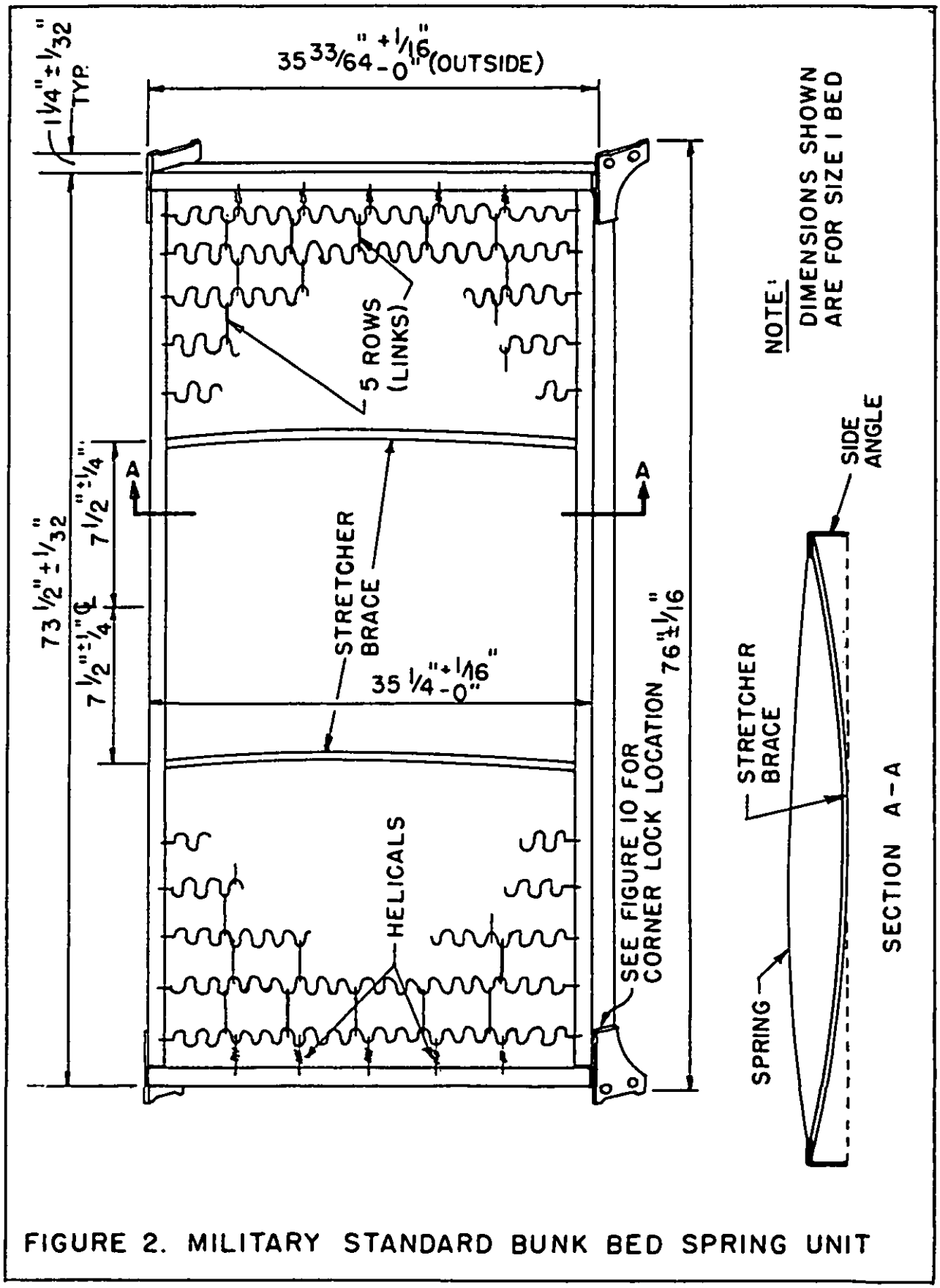


FIGURE 2. MILITARY STANDARD BUNK BED SPRING UNIT

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

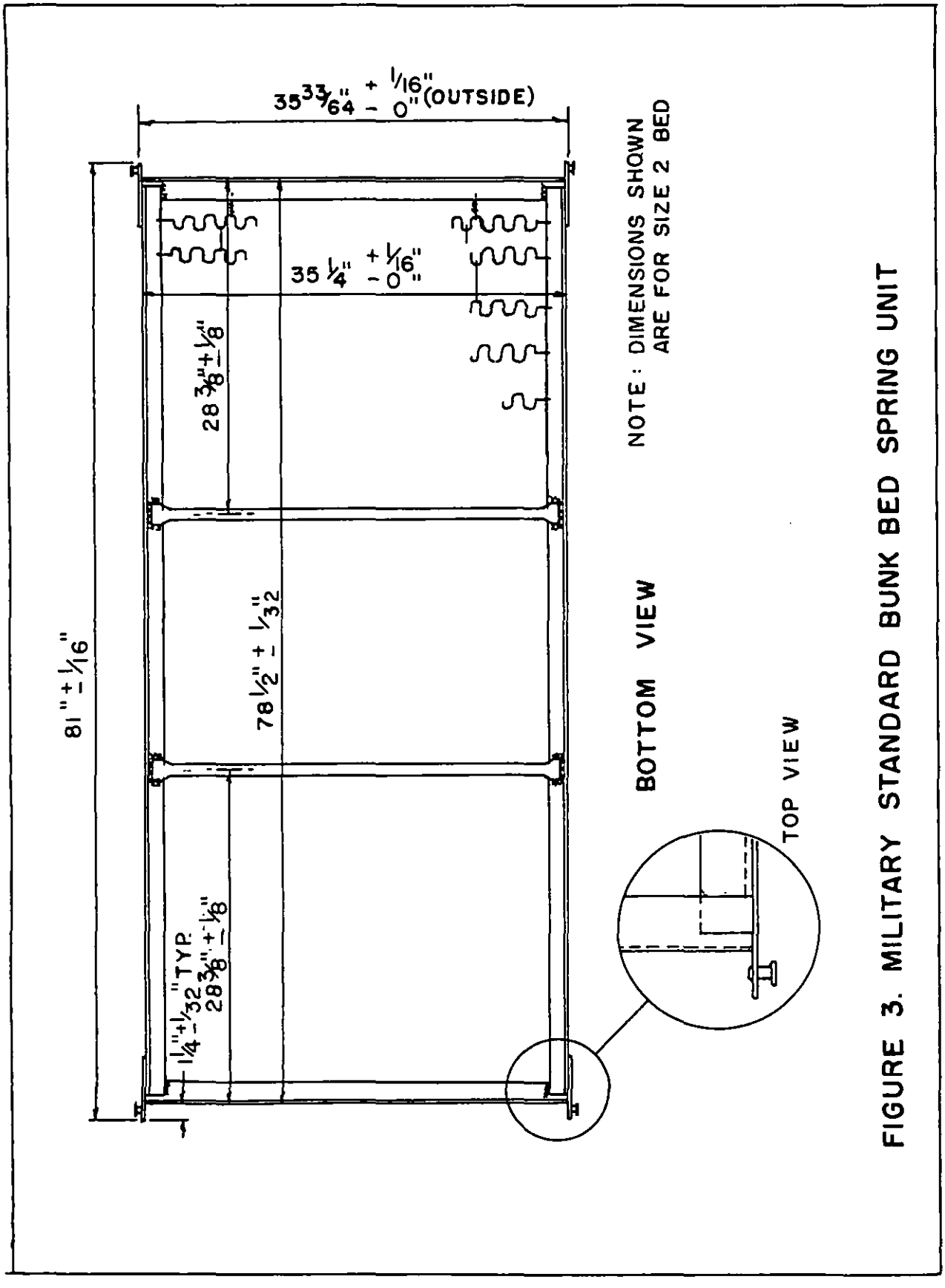
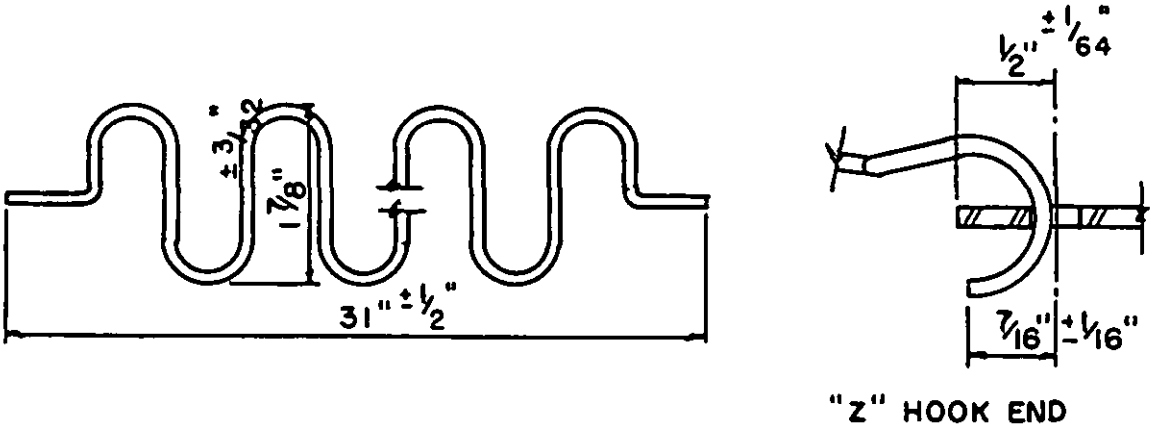
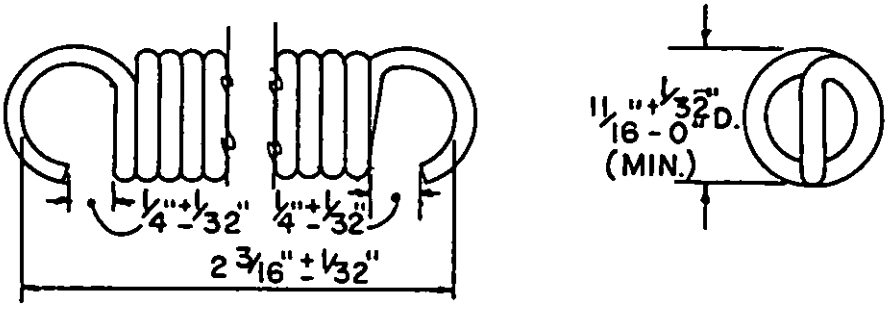


FIGURE 3. MILITARY STANDARD BUNK BED SPRING UNIT

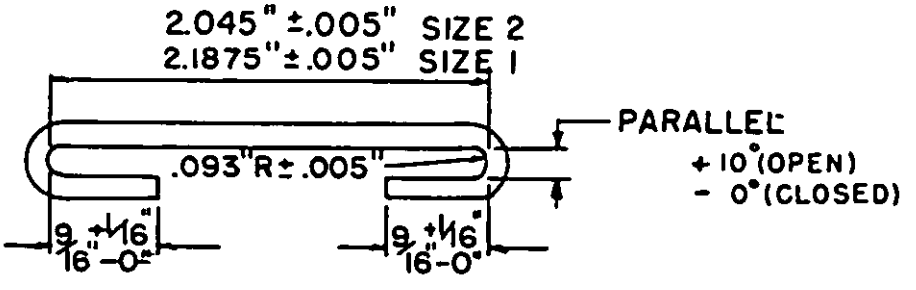
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.



A. SPRING FABRIC (no sag type)



B. SPRING FABRIC HELICALS
(12 turns close wound)



C. SPRING FABRIC LINK

FIGURE 4. SPRING UNIT COMPONENTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

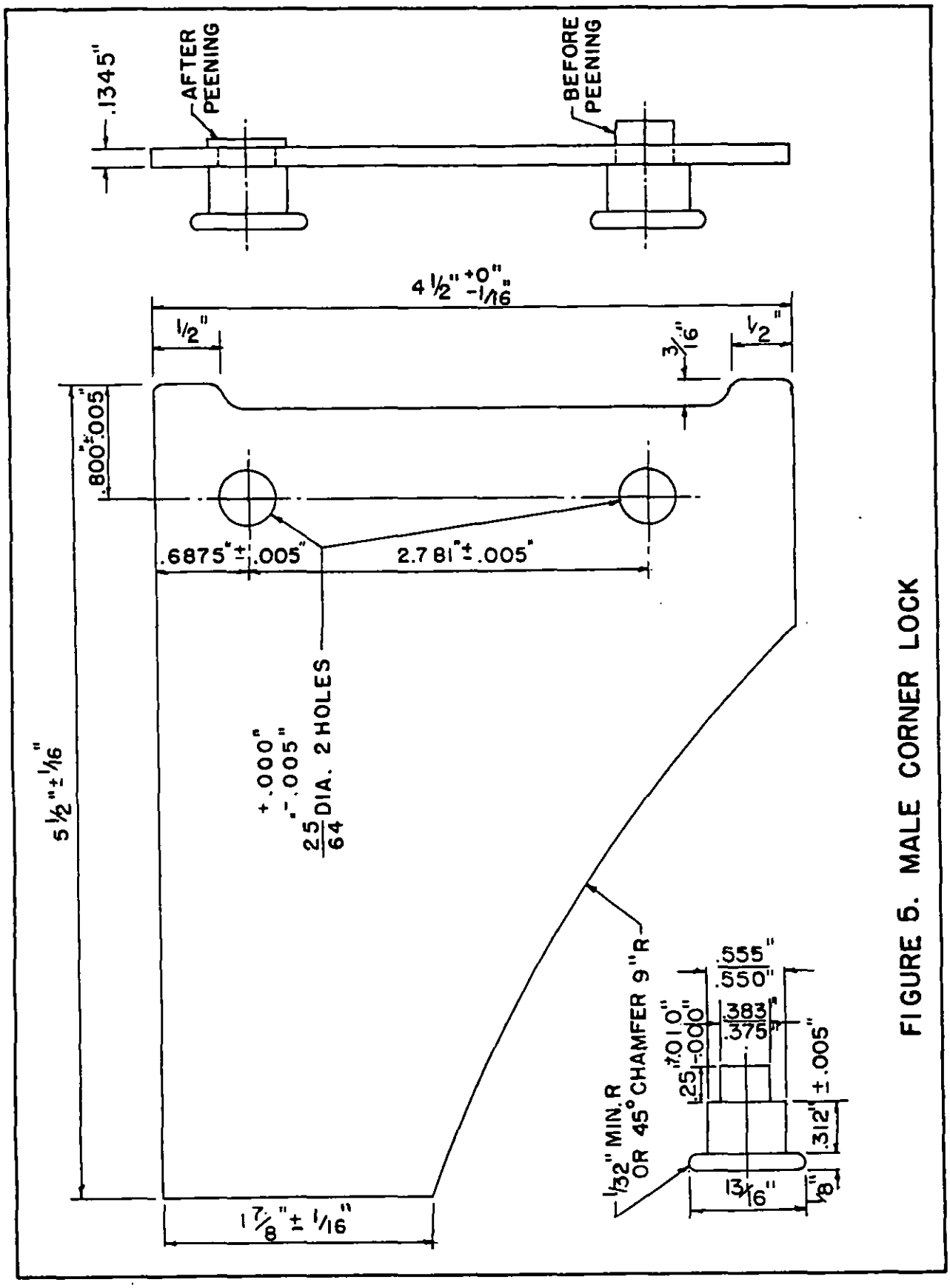


FIGURE 6. MALE CORNER LOCK

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

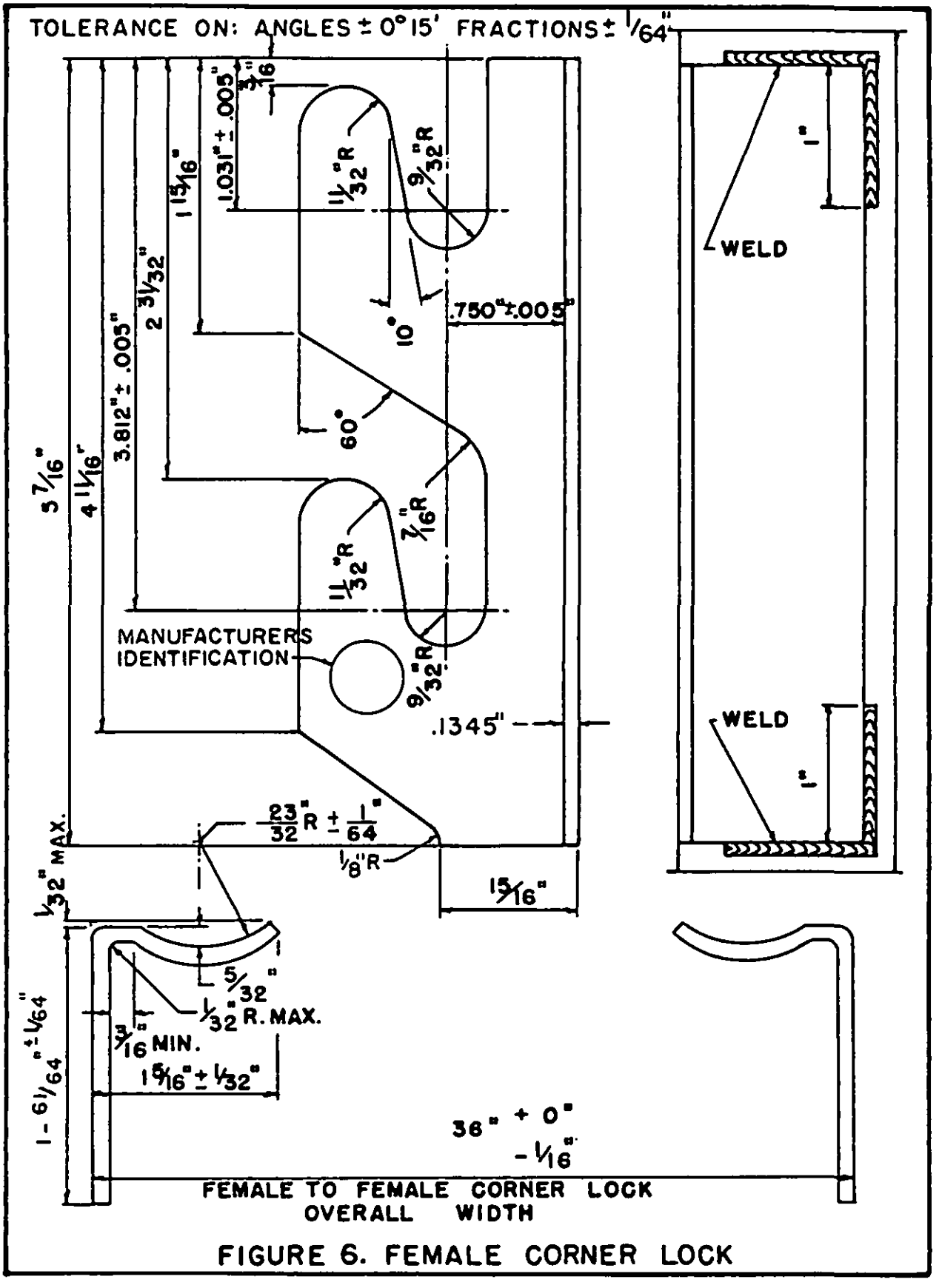


FIGURE 6. FEMALE CORNER LOCK

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

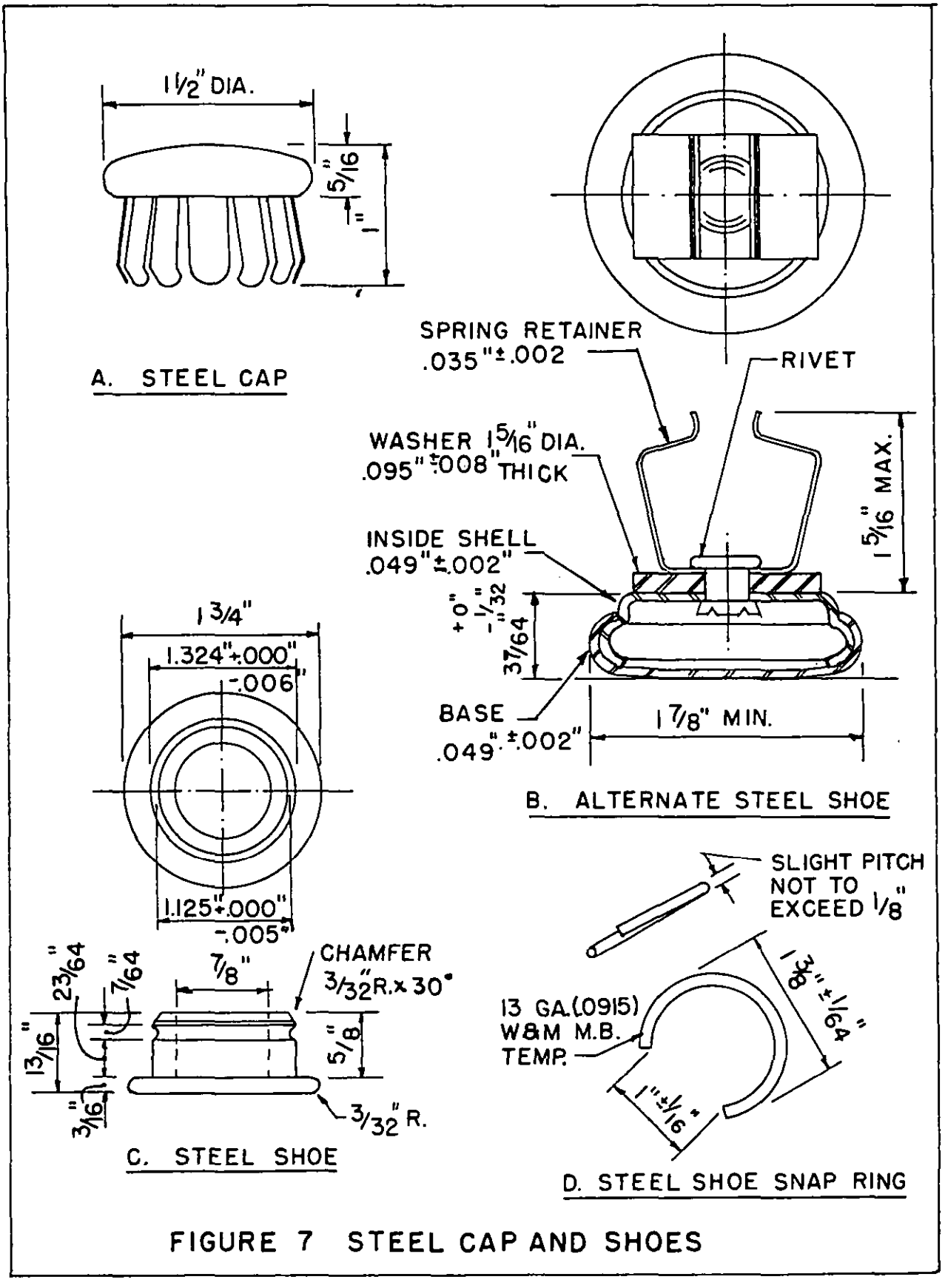


FIGURE 7 STEEL CAP AND SHOES

Source: https://assist.dla.mil/ -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

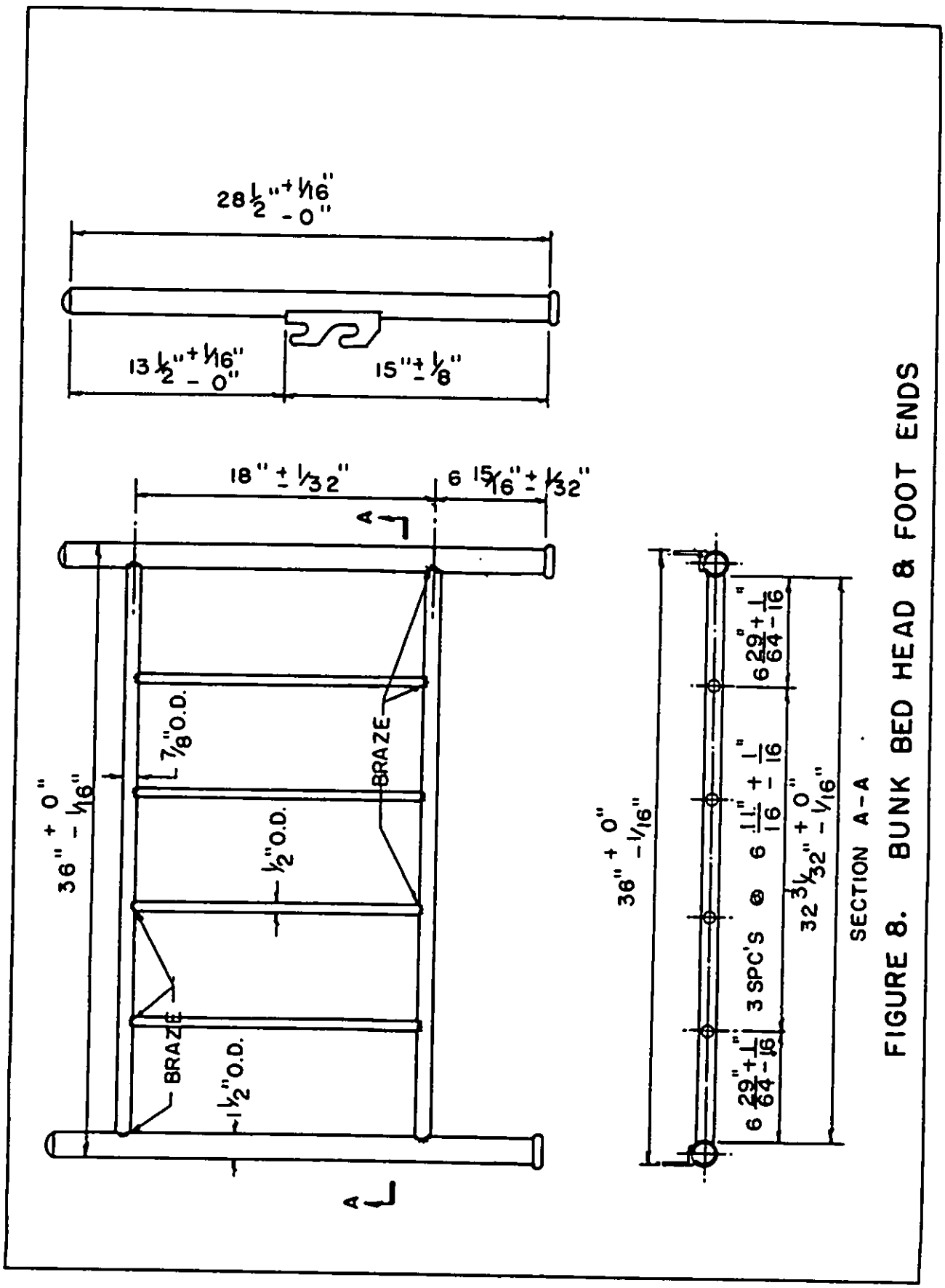


FIGURE 8. BUNK BED HEAD & FOOT ENDS

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

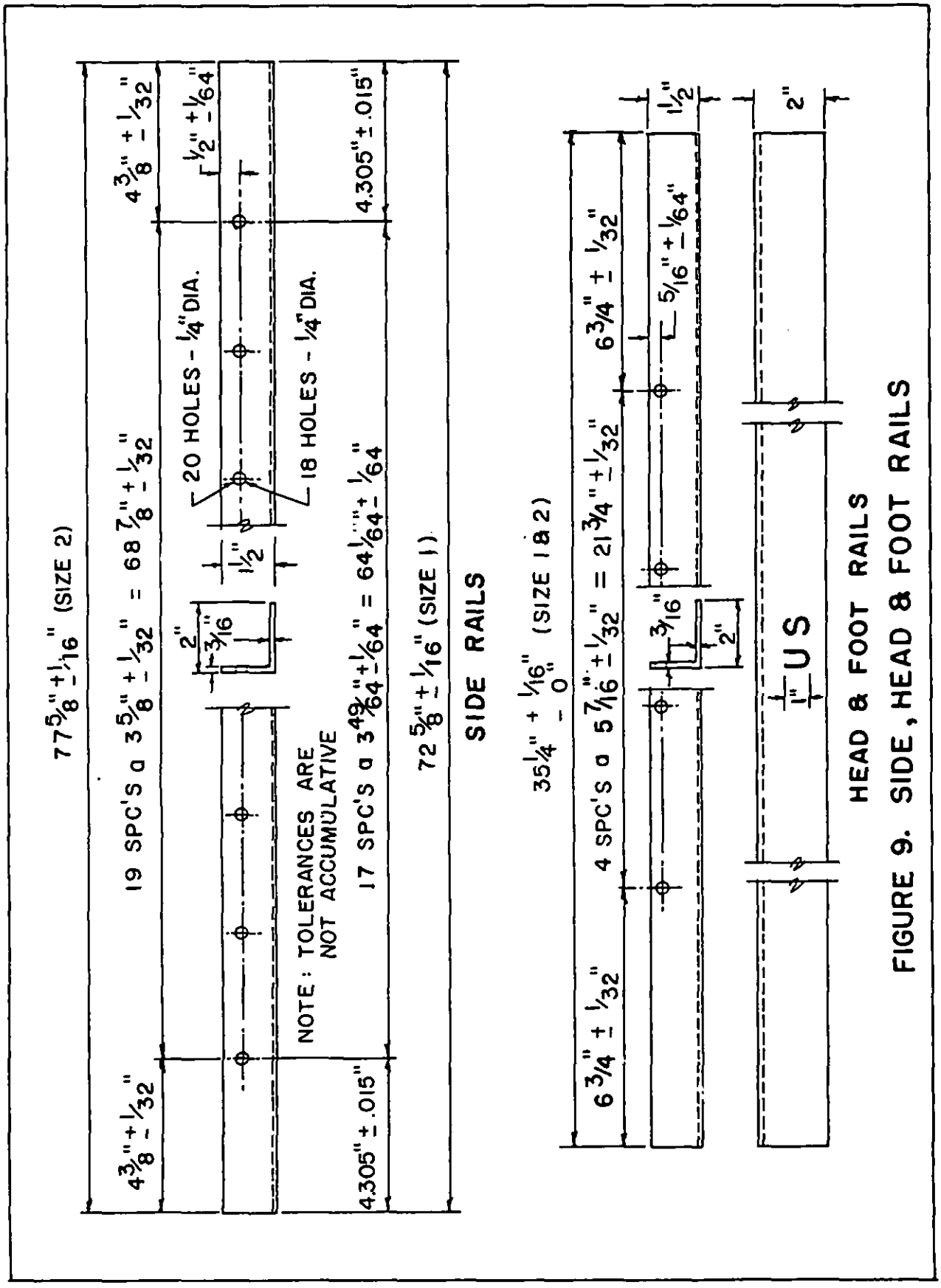


FIGURE 9. SIDE, HEAD & FOOT RAILS

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

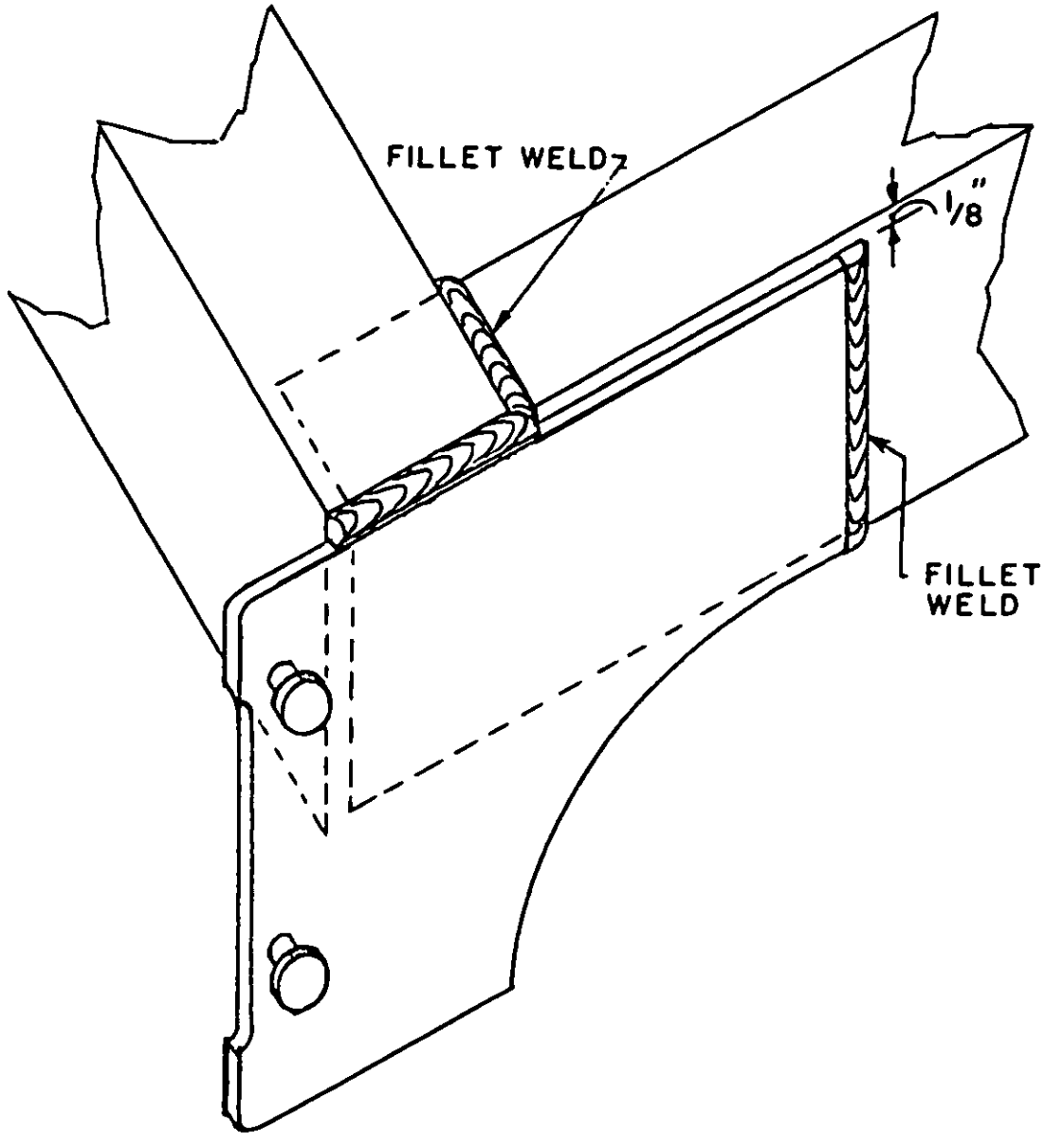


FIGURE 10. SPRING FRAME WELDING, TOP VIEW .....

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

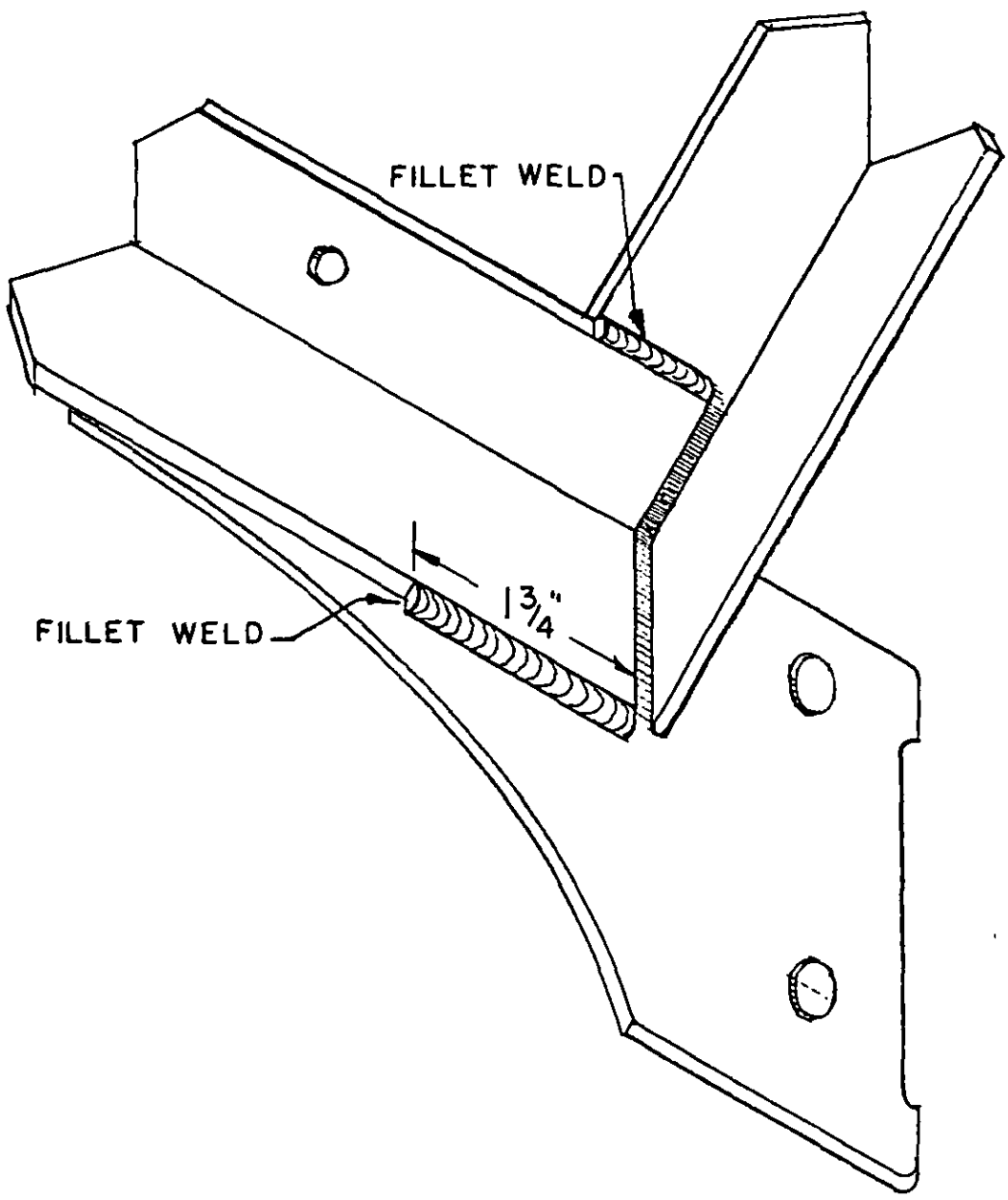


FIGURE 11. SPRING FRAME WELDING, BOTTOM VIEW .....

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

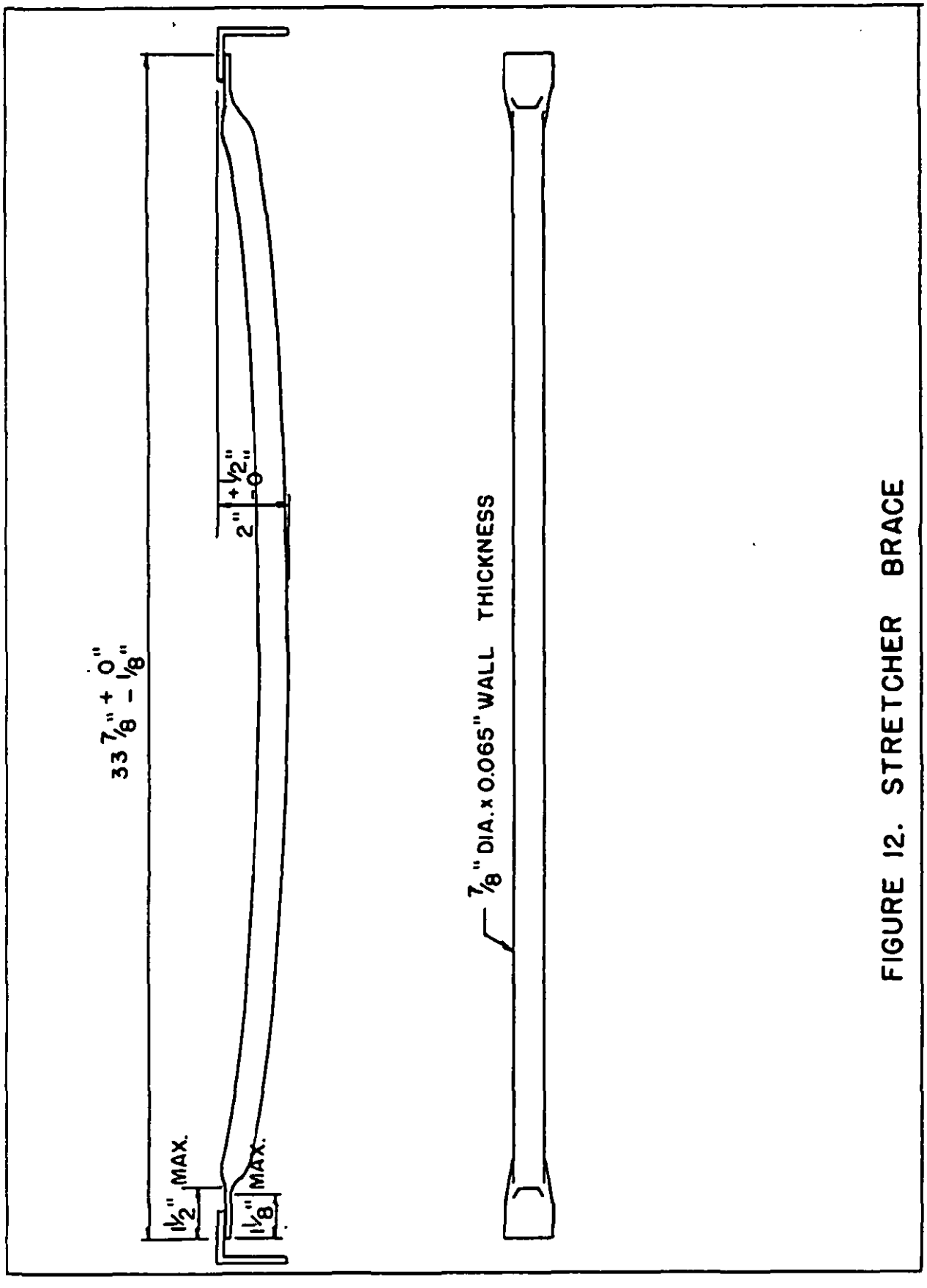


FIGURE 12. STRETCHER BRACE

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

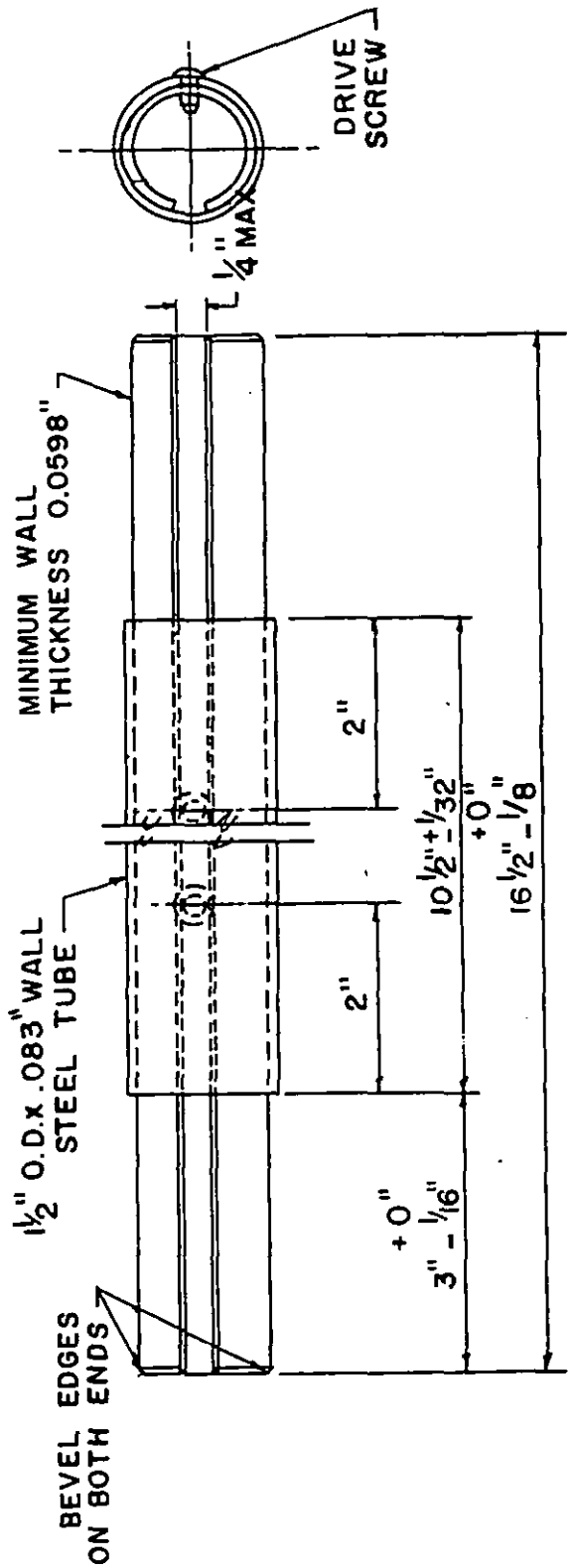


FIGURE 13. ADAPTER FOR CONVERSION FROM SINGLE TO DOUBLE-DECK BED

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

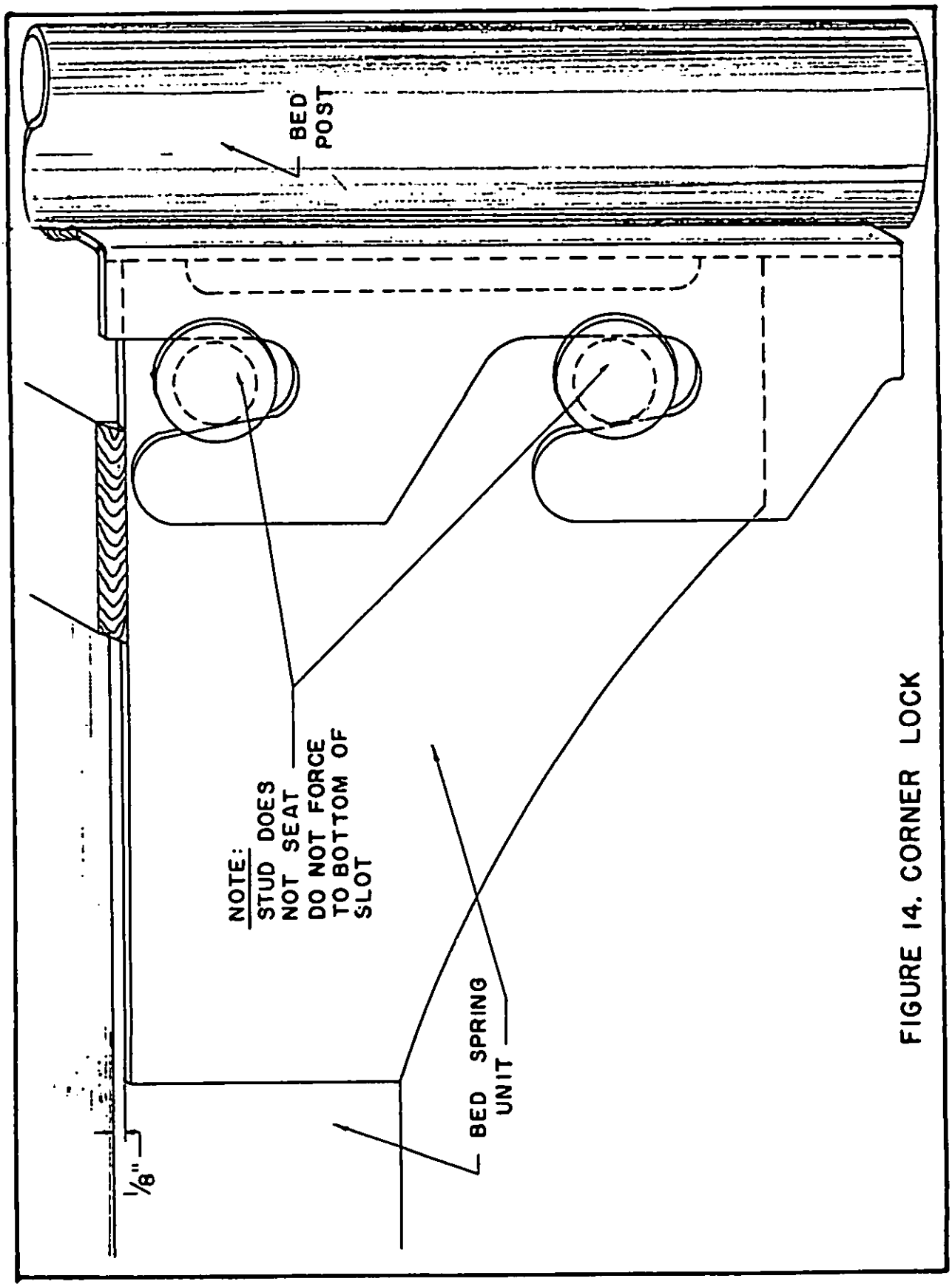


FIGURE 14. CORNER LOCK

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-15228F BED, BUNK, STEEL, ROUND TUBE, SINGLE AND DOUBLE-DECK (CONVERTIBLE TYPE)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Facilities Engineering Command
200 Stovall Street
Alexandria, VA 22332

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:54Z
Check the source to verify that this is the current version before use.